

Nila Mönig, PhD

moenigin@gmail.com | Freiburg (DE) | German | linkedin.com/in/nilamoenig | github.com/moenigin

PROFILE

PhD neuroscientist turning large, complex image datasets into reliable, reproducible results. Develops deep-learning pipelines (PyTorch, HPC) together with the tools, quality control and documentation around them, and works across biology, engineering and external partners. Hands-on from hardware to code: builds solutions where no standard tool exists.

CORE COMPETENCES

Computational microscopy: Electron microscopy | Segmentation | Object classification | Feature extraction

Machine learning & AI: PyTorch | MONAI | Gunpowder | scikit-learn | U-Net / ResNet architectures

Programming & tools: Python | MATLAB | Git | napari | Neuroglancer | FIJI/ImageJ | ilastik

Workflow & reproducibility: Automation | Versioned pipelines | Documentation | Quality control | Benchmarking

Scalable data processing: SLURM/HPC | Array-based deployment | Parallel inference | Process automation

Data analytics: Statistical analysis | Multi-dimensional data analysis | Clustering | Dimensionality reduction

Hardware and prototyping: Prototyping | Fabrication | Fluidics | Control electronics | Soldering

Collaboration & communication: Cross-disciplinary collaboration | Partner coordination | Mentoring and knowledge transfer | Scientific writing | Presentation

PROFESSIONAL EXPERIENCE

Independent Work

Jul 2026 - present

- Finalising the synapse-detection pipeline (code cleanup, documentation, validation) for release
- Revising a manuscript for journal publication

POSTDOCTORAL SCIENTIST

Friedrich Miescher Institute for Biomedical Research, Basel - Oct 2022 - Jun 2026

Led development of scalable data pipelines for connectomics analysis in zebrafish neuroscience

- **Designed and implemented** a high-throughput synapse-detection pipeline tailored to non-standard tissue properties.
- **Developed, trained, and deployed** deep-learning segmentation and classification models (U-Net, ResNet variants) on HPC infrastructure using array-based parallel deployment.
- **Tool development:** created and maintained custom proofreading and training-data tools in napari and Neuroglancer, with documentation and user feedback integrated; collaborated with imaging-facility specialists on tool development.
- **Partner coordination:** set up and coordinated a high-throughput proofreading partnership with an external company; established benchmarking protocols, ran quality control and wrote interface scripts to reconcile proofreading platforms and data formats.
- **Training data:** generated ground-truth training data for segmentation of a highly complex dataset, drawing on extensive experience with volume EM.
- **Strategy and data quality:** designed the experimental and computational strategy for the connectomics component; evaluated image quality and gave feedback to acquisition teams to support segmentation readiness across datasets of different origins.
- **Knowledge transfer:** advised colleagues informally on ground-truth generation and introduced them to connectomics methods and terminology; supervised an intern in data annotation and tool use.
- **Documentation and writing:** wrote documentation for custom tools and a detailed annotation guideline defining the criteria for synapse annotation.

PHD CANDIDATE

Friedrich Miescher Institute for Biomedical Research - May 2012 - Sep 2022

Thesis: Neuronal Organisation of the Olfactory Bulb in Adult Zebrafish | magna cum laude (5.5/6.0)

Developed integrated connectomics and behavioural-analysis workflows to investigate neural-circuit function in the zebrafish olfactory system.

- **Collaboration with Google Research:** partnered on neuron segmentation for TB-scale EM data; assessed model output by visual inspection, reported errors and generated targeted ground-truth data to close gaps.
- **Ground truth and data quality:** generated ground-truth training datasets, gaining a deep understanding of computer-vision model requirements and data-quality standards.
- **Proofreading tools:** developed a Neuroglancer-based interface for inspecting and curating automated segmentations (15–50× faster reconstruction) and a tool for manual neuron classification.
- **Analysis:** proofread >450 neurons and established a morphological classification scheme.
- **Process automation:** co-developed and automated behavioural assays for odour discrimination in adult zebrafish, cutting manual work by 87% while doubling throughput. Implemented a multi-dimensional analysis pipeline (Python/MATLAB).
- **Hardware:** designed and built custom behavioural apparatus (prototyping, fabrication, fluidics, control electronics).
- **Cross-disciplinary projects:** contributed behavioural training and analysis to projects linking neural function and behaviour.
- **Mentoring:** trained a student assistant in neuron proofreading.

RESEARCH ASSISTANT, University of Freiburg – Jan 2011 – Mar 2012

- **Prepared and maintained** primary neuronal cell cultures ensuring viability and experimental quality.
- **Implemented** closed-loop stimulation circuits (Tucker Davis Technologies).
- **Performed** electrophysiology, pharmacology, immunohistochemistry and neuron tracing.

STUDENT ASSISTANT, University of Freiburg – Jun 2007 – Sep 2009

- **Teaching assistant** for introductory MATLAB course.
 - **Contributed** to data-management system development and neuroanatomical analyses.
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EDUCATION

- PhD Neuroscience, University of Basel (2022) – magna cum laude
- Diplom Biology (≈ MSc), University of Freiburg (2009) – 1.0 (best possible)

SELECTED PUBLICATIONS & PRESENTATIONS

- Mönig N.R*, Januszewski M.* et al., 2026, **bioRxiv** 2026.07.13.738197 (preprint)
- Frank T., Mönig N.R., Satou C. et al., **Nat Neurosci** 22: 1844–1856 (2019)
- Namekawa I.*, Mönig N.R*, Friedrich R.W. **Exp Brain Res** 236: 2959–2969 (2018)

(*equal contribution)

Poster presentation at international meetings (CONNECTOMICS, FENS, ECRO, ZEBRAFISH BRAIN CONFERENCE). Talks at regional meetings.

SELECTED COURSES & TRAINING

- Entrepreneurship Training, Innosuisse Module 2 (Innovation Office, University of Basel, 2026)
- Advanced Python and Machine Learning (University of Basel, 2024)
- Dealing with Research Data Management (University of Basel, 2024)
- Deep Learning for image analysis (EMBL, Heidelberg, 2023)
- antelope@novartis (University of Basel, 2017)

LANGUAGES

German (native), English (C2), French (B1)

INTERESTS

Volunteer conservation actions (reforestation, nature-reserve clean-ups), running, furniture upcycling, knitting